

AHMAD PEYVAN

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EDUCATION

University of Illinois at Chicago, USA

PhD in Mechanical Engineering

GPA: 4.0/4.0

Sharif University of Technology, Iran

MS in Mechanical Engineering-Energy Conversion

GPA: 3.69/4.0

Shiraz University, Iran

BS in Mechanical Engineering-Mechanics in Heat and Fluids

GPA: 3.65/4.0

WORK EXPERIENCE

Brown University

August 2024 - December 2025

Assistant Professor of Applied Mathematics (Research)

Providence, RI

- Conduct research on a virtual laboratory framework using large language model-driven multi-agent paradigms for aerial vehicle geometric design.
- Advise a PhD student on developing reduced-order dynamical models for supersonic and hypersonic intakes using physics-informed neural networks (PINNs).
- Advise undergraduate student on robust inverse design of electromagnetic devices using uncertainty-quantified neural operators and CVaR optimization.
- Geometry-dependent neural operator for hypersonic flow prediction on arbitrary grids with limited data.
- Develop a high-order spectral element method solver for the simulation of hypersonic flows with high parallel efficiency, adaptive mesh refinement, and realistic thermophysical modeling.

PredictiveIQ

December 2023 - August 2025

Artificial Intelligence Specialist

Providence, RI

- Shape optimization of industrial evaporators to maximize heat transfer.
- Create surrogate models based on Deep Operator Networks for heat transfer of boiling refrigerants inside evaporators coil.
- Create a deep learning-based surrogate model to predict the incompressible flow field around Scania trucks, aimed at optimizing the cabin's shape to reduce drag forces.
- General geometry parameterization of 3D point clouds using deep neural networks (PointNet).

Brown University

January 2022 - July 2024

Postdoctoral Research Associate

Providence, RI

- Develop entropy stable discontinuous Galerkin spectral element method in *Julia* for inviscid hypersonic flows.
- Develop surrogate models based on deep neural operators to infer the flow field around an airfoil and its geometric optimization.
- DNS of laminar to turbulent transition in hypersonic boundary layer using a *Fortran* code.
- Train deep neural operators to learn the solution of Riemann problems using *JAX*.

University of Illinois as Chicago

January 2017 - December 2022

Research Assistant

Chicago, IL

- Develop a three-dimensional compressible flow solver (discontinuous spectral element method) in *MPI-Fortran* to perform large-scale parallel simulations (using 130,000 processors).

- Develop a new high-order numerical method called flux reconstruction discontinuous spectral element method (FRDSEM) for solving compressible Navier-Stokes's equations.
- Develop double flux and shock-capturing techniques compatible with FRDSEM to simulate multicomponent compressible turbulent flows inside shock tubes.
- Conduct numerical simulation of COVID-19 virus propagation inside a dentistry room in a dentistry clinic using *ANSYS/Fluent*.

Industrial Assessment Center

January 2017 - December 2022

Intern

Chicago, IL

- Conduct research for energy efficiency measures.
- Perform energy efficiency assessments for small to medium-sized plants in the Midwest under the Department of Energy grant.
- Identify more than \$130,000 in potential annual energy (Natural Gas, Electricity, and Water) saving opportunities in small to medium-sized plants.

Sharif University of Technology

September 2011 - September 2013

Research Assistant

Tehran, Iran

- Develop a *MATLAB* code for axial flow compressor designing.
- Conduct numerical research on the performance of axial flow compressors using *ANSYS/CFX*.
- Test turbojet engines to acquire performance curves of their main components, such as axial compressors.

GRANTS

- PI: Supercomputing allocation grant from the U.S. National Science Foundation ACCESS titled "Large Eddy Simulations of Chemically Reacting Turbulent Flow."
- Co-PI: Supercomputing allocation grant from the National Center for Supercomputing Applications titled "DNS Investigation of Compressibility Effects on Separating and Reattaching Shear Layers."
- Writing: DURIP-GPU Cluster for Neural PDEs and Neural Operators to Support MURI Research and Beyond: US\$ 750,000, Start Date: 02/01/2024, End Date: 01/31/2025.
- Writing: MURI ADD-ON: Co-writing: Physics-Informed neural networks for high speed flows (AFOSR): US\$ 300,000
- Writing: SMASH project: Heat transfer on hypersonic vehicles (ARL): US\$ 200,000.

PUBLICATIONS: GOOGLE SCHOLAR CITATION: 420, H-INDEX: 12

Journal Publications

1. **A. Peyvan**, K. Shukla, J. Chan, and G. Karniadakis, High-order methods for hypersonic flows with strong shocks and real chemistry. *Journal of Computational Physics* (2023): 112310. [Link](#).
2. **A. Peyvan**, K. Shukla, and G. Karniadakis, H3PC: Hypersonic, High-Order, High-Performance Code with Adaptive Mesh Refinement and Real Chemistry. arXiv preprint arXiv:2511.17551 (2025). [Link](#)
3. K. Shukla, V. Oomen, **A. Peyvan**, ..., G. Karniadakis, Deep neural operators as accurate surrogates for shape optimization. *Engineering Applications of Artificial Intelligence* (2024) [Link](#)
4. **A. Peyvan**, A. Jagtap, G. Karniadakis, RiemannOnets: RiemannONets: Interpretable neural operators for Riemann problems *Computer Methods in Applied Mechanics and Engineering*, (2024) 426, p.116996 [Link](#)
5. **A. Peyvan**, V. Kumar, G. Karniadakis, Fusion-DeepONet: A Data-Efficient Neural Operator for Geometry-Dependent Hypersonic and Supersonic Flows. *Journal of Computational Physics*, (2025) p114432 [Link](#)

6. B. Shih, **A. Peyvan**, Z. Zhang, G. Karniadakis, Transformers as Neural Operators for Solutions of Differential Equations with Finite Regularity. *Computer Methods in Applied Mechanics and Engineering*, (2025) 434. [Link](#)
7. **A. Peyvan**, L. Bravo, A. Ghosal, O. Marxen, and G. Karniadakis, Hypersonic boundary layer transition and heat loading. *arXiv preprint arXiv:2407.02311* (2024). [Link](#).
8. E. Roohi, A. Shoja-Sani, B. Goshayeshi, **A. Peyvan**, Learning Rarefied Gas Dynamics with Physics-Enforced Neural Networks. *arXiv preprint arXiv:2509.06231* (2025). [Link](#).
9. E. Kiani, V. Ananchaperumal, **A. Peyvan**, ... , Crack Path Prediction with Operator Learning using Discrete Particle System data Generation. *arXiv preprint arXiv:2506.01976* (2025). [Link](#).
10. O. Ovadia, V. Oommen, A. Kahana, **A. Peyvan**, ... , Real-time inference and extrapolation with Time-Conditioned UNet: Applications in hypersonic flows, incompressible flows, and global temperature forecasting *Computer Methods in Applied Mechanics and Engineering* 441 (2025): 117982. [Link](#).
11. **A. Peyvan**, D. Li, J. Komperda, F. Mashayek, Oscillation-free nodal discontinuous spectral element method for the simulation of compressible multicomponent flows. *Journal of Computational Physics*, (2022) 452, p.110921. [Link](#)
12. D. Li, J. Komperda, **A. Peyvan**, Z. Ghiasi, F. Mashayek. Assessment of turbulence models using DNS data of compressible plane free shear layer flow. *Journal of Fluid Mechanics* 931 (2022): A10. [Link](#)
13. **A. Peyvan** , J. Komperda, D. Li, Z. Ghiasi, F. Mashayek. Flux reconstruction using Jacobi correction functions in discontinuous spectral element method. *Journal of Computational Physics* (2021): 110261. [Link](#)
14. D. Li, **A. Peyvan**, Z. Ghiasi, J. Komperda, F. Mashayek. Compressibility effects on energy exchange mechanisms in a spatially developing plane free shear layer. *Journal of Fluid Mechanics* 910 (2021). [Link](#)
15. J. Komperda, **A. Peyvan**, ..., F. Mashayek, Computer simulation of the SARS-CoV-2 contamination risk in a large dental clinic. *Physics of Fluids* 33, 033328 (2021)(Featured paper) [Link](#)
16. Z. Ghiasi, J. Komperda, D. Li, **A. Peyvan**, D. Nicholls, F. Mashayek, Modal explicit filtering for large eddy simulation in discontinuous spectral element method. *Journal of Computational Physics: X* 3 (2019): 100024. [Link](#)
17. D. Li, J. Komperda, Z. Ghiasi, **A. Peyvan**, F. Mashayek, Compressibility effects on the transition to turbulence in a spatially developing plane free shear layer. *Theoretical and Computational Fluid Dynamics* 33, no. 6 (2019): 577-602. [Link](#)
18. **A. Peyvan**, A. H. Benisi. Axial-Flow Compressor Performance Prediction in Design and Off-Design Conditions through 1-D and 3-D Modeling and Experimental Study. *Journal of Applied Fluid Mechanics* 9, no. 5 (2016). [Link](#)

Refereed conference proceedings

1. **A. Peyvan** Neural Operator-based modeling of hypersonic flows around varying geometries, Digital Twins in Engineering and Artificial Intelligence and Computational Methods in Applied Science (2025).
2. **A. Peyvan**, G. Karniadakis, Solution of Riemann Problems with Real Chemistry for Compressible Flows Using Neural Operators, 16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics (2024).
3. **A. Peyvan**, D. Li, J. Komperda, F. Mashayek, Large-Scale Simulations of Turbulence Using Discontinuous Spectral Element Method. World Academy of Science, Engineering and Technology,

International Science Index, Mathematical and Computational Sciences (2018), 12(8), 1264.

4. **A. Peyvan**, D. Li, J. Komperda, Z. Ghiasi, F. Mashayek, Under-resolved Simulation of Backward Facing Step Flow Using Flux Reconstruction Discontinuous Spectral Element Method. In AIAA SCITECH 2022 Forum, p. 0486. 2022.
5. J. Komperda, D. Li, **A. Peyvan**, F. Mashayek. Filtered Density Function for Shocked Compressible Flows on Unstructured Spectral Element Grids. In AIAA Scitech 2020 Forum, p. 1789. 2020.

INVITED PRESENTATIONS

1. Invited to the Department of Mathematics, Massachusetts Institute of Technology, USA. **Title of talk: Modeling High-Speed Flows: Approaches for Hypersonic Flows with Strong Shocks, Real Chemistry, and Interpretable Neural Operators for Riemann Problems**
2. Invited to the Department of Computational and Applied Mathematics, Rice University, USA. **Title of talk: High Order Methods for Hypersonic Flow Simulations**
3. Invited to the Department of Applied Mathematics, Brown University, USA. **Title of talk: Direct Numerical Simulation of Hypersonic Boundary Layer Transition Over a Flat Plate.**

HONORS AND AWARDS

1. Fellowship award from College of Engineering, University of Illinois at Chicago, Spring 2017.
2. Selected as a featured article in Physic of Fluid Journal for "Computer simulation of the SARS-CoV-2 contamination risk in a large dental clinic" paper.
3. Industrial Assessment Center (IAC) certificate for exceptional participation in the IAC program under the US Department of Energy.
4. Top ten download list for "High-order methods for hypersonic flows with strong shocks and real chemistry" article in Chemical Physics eJournal, listed on SSRN's.
5. Ranked 1st GPA among B.Sc. students' of 2007 class, School of Mechanical Engineering, Shiraz University, Shiraz, Iran.
6. Ranked 6th among over 60 elites in the Mechanical Engineering Students Olympiad, Iran, 2011.
7. Ranked 10th among more than 1,700 participants in the Ph.D. program University entrance exam, Iran, 2014.

SKILLS

Programming languages: FORTRAN, Matlab, Julia, C++, Python, Maple, Shell Scripting.

Parallel programming models: MPI, OpenMP

Machine learning packages : JAX, PyTorch, TensorFlow

CFD software: ANSYS/Fluent, ANSYS/CFX, OpenFOAM

Mesh generation for CFD: GridPro, Fluent/Meshing module, ANSYS meshing module, Trelis, Gambit

Post Processing software: ParaView, VisIt, Tecplot, Python matplotlib, MATLAB

Designing software: SolidWorks

TEACHING

Sharif University of Technology

- Teaching assistance for the **Gas Turbine** course for undergraduate students.

University of Illinois at Chicago

- Teaching assistance for the **Air Pollution Control** course for graduate students.

- Teaching assistance for the **Introduction to Thermodynamics** course for undergraduate students.

COURSES TAKEN

Mechanical Engineering

- Fuel Cell systems
- Aircraft principle
- Advanced Turbomachinery
- Computational Compressible Flow
- Fundamental of Turbulence

Computational and Applied Mathematics

- Deep Learning for Scientists and Engineers

PROFESSIONAL AFFILIATION

- American Institute of Aeronautics and Astronautics.
- American Physical Society, Division of Fluid Dynamics.

JOURNAL REVIEWERS

- Journal of Computational Physics
- Physics of Fluids
- Computer Methods in Applied Mechanics and Engineering
- Journal of Fluid Mechanics

FIELDS OF INTEREST

- Hypersonics
- Deep neural operators
- Physics-informed neural networks (PINNs)
- Turbulent reactive flows
- Multiphase flows
- High order numerical methods
- Large-scale turbulent flow simulations on HPCs
- Parallel programming
- Turbo-machinery